

Data Visualization – Lec/Lab

Lesson 7 - Visualizing Relationships and Correlations

Name: ____

Assignment: (Individual)

Task 1:

Summarize the core concept of the following subtopics:

- A. Scatter Plots: The primary tool for visualizing the relationship between two quantitative variables.
- B. Bubble Charts: An extension of scatter plots that adds a third dimension (size) to represent a third variable.
- C. Correlation Heatmaps: Using color-coded grids to summarize relationships across a matrix of multiple variables simultaneously.
- D. The Pearson Coefficient (ρ or r): Understanding the mathematical score (from -1 to +1) that defines the strength and direction of a linear relationship.
- E. Regression Analysis: Introduction to "Best Fit" lines used to predict future outcomes based on current correlations.

Deliverables:

2-3 pages in a PDF file.

Task 2:

Calculating the Pearson Correlation (r):

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

Components of the Formula

- r : Pearson Correlation Coefficient
- n : Number of pairs of data
- $\sum xy$: Sum of the product of paired scores
- $\sum x$: Sum of x scores
- $\sum y$: Sum of y scores
- $\sum x^2$: Sum of squared x scores
- $\sum y^2$: Sum of squared y scores  Britannica +4

Task Description:

Date: March 30 2026

You are analyzing the relationship between Study Hours (x) and Exam Scores (y) for five students. You must manually calculate the Pearson Coefficient to determine if more study time actually results in higher scores.

Data Table:

Student	Study Hours (x)	Exam Score (y)
A	2	70
B	4	75
C	6	85
D	8	90
E	10	95

Required Deliverables:

1. Complete a calculation table for x^2 , y^2 , and xy .
2. Calculate the Pearson r using the formula.
3. Interpret the result: Is it a weak, moderate, or strong correlation?

Task 3: Regression Analysis & Prediction:

Task Description: A coffee shop wants to predict Daily Revenue (y) based on the Number of Customers (x). You are given the "Line of Best Fit" equation derived from their last month of data.

The Equation: $y = 15.5x + 120$

(Where 15.5 is the average spend per customer and 120 represents fixed base revenue like catering orders).

Data for Validation:

Day	Customers (x)	Actual Revenue (y)
Monday	40	\$750
Tuesday	60	\$1,050

Date: March 30 2026

Required Deliverables:

1. Residual Analysis: Calculate the "Predicted Revenue" for Monday and Tuesday using the equation. Then, find the Residual (Actual Revenue minus Predicted Revenue).
2. The Prediction: If the shop expects 100 customers on Saturday, what is the predicted revenue?
3. Slope Interpretation: What does the number 15.5 represent in real-world business terms?

Note: Output must be in a PDF file.

Due date: April 6, 2026